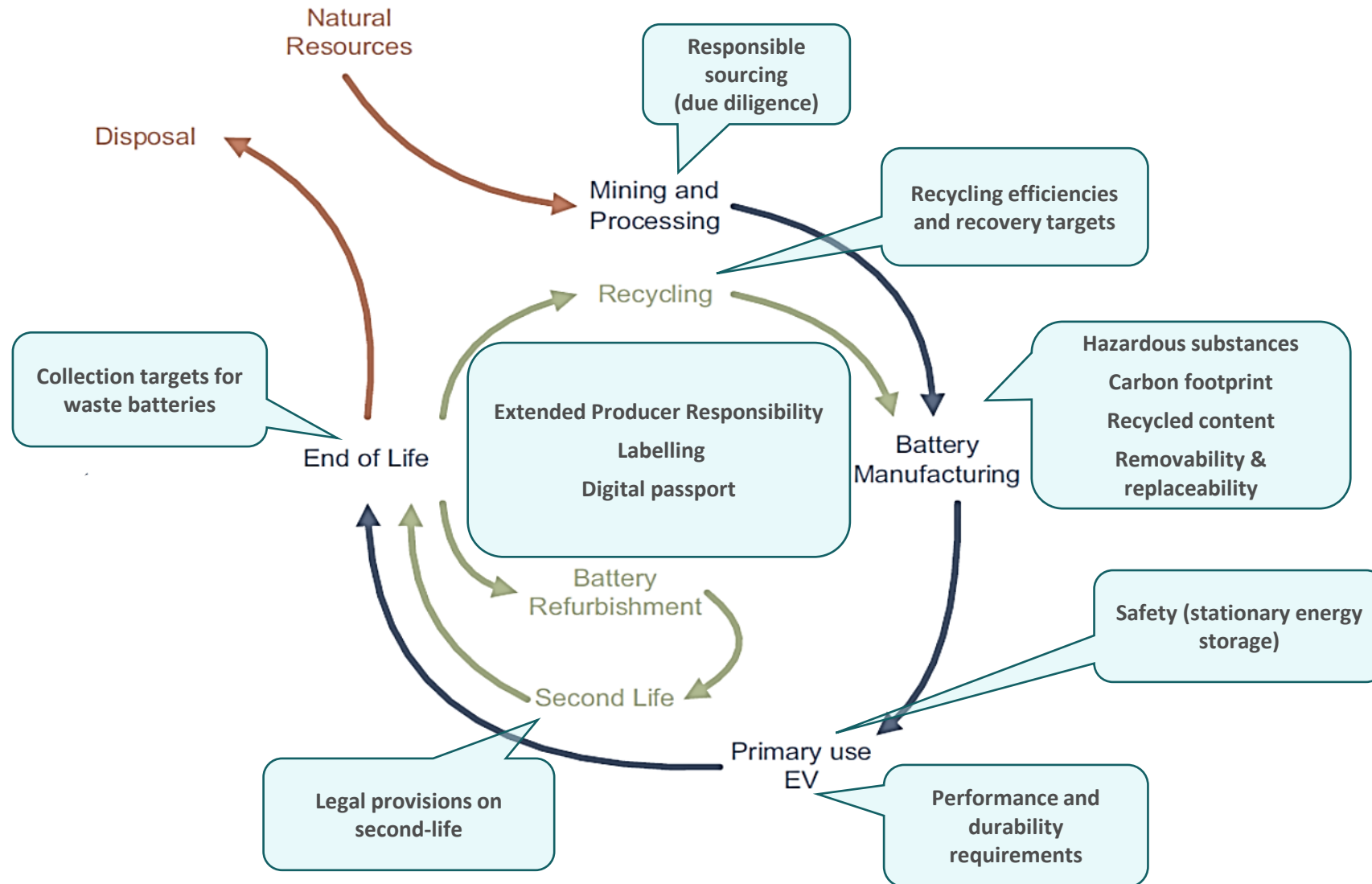




Regulation (EU) 2023/1542 on batteries and waste batteries

GRPE hybrid workshop, 17/10/2024

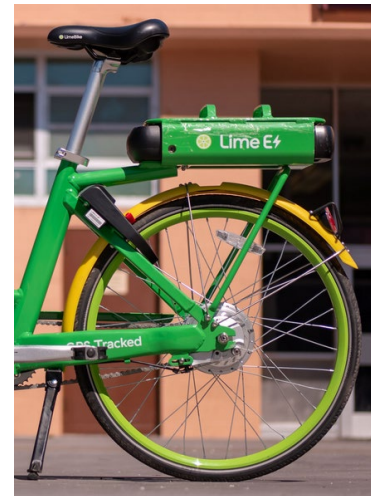
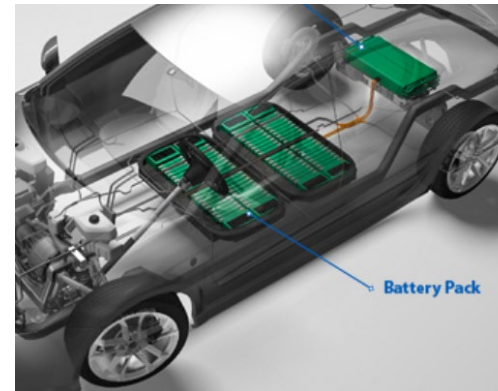
Covering the entire life cycle:



Classification of batteries

Five categories, with some subcategories

- **portable battery (up to 5 kg, not for industrial use), incl.:**
 - portable batteries of general use: common formats AA, AAA, 9V, ..
- **starter, lighting or ignition (SLI) battery**
- **light means of transport (LMT) battery**
- **electric vehicle (EV) battery**
- **industrial battery, incl.:**
 - stationary battery energy storage systems
 - flow batteries ('with external storage')



Requirements for placing on the EU market

- Design of the battery: restriction of hazardous substances, performance & durability, safety
- Production of the battery: carbon footprint, recycled content
- Providing information on the battery: marking & labelling, battery passport, battery management system
- On economic operators: due diligence (address social and environmental risks in sourcing and processing of lithium, natural graphite, cobalt & nickel)

Certain requirements apply only to specific categories of batteries, e.g. for reasons of proportionality, or if already addressed by other legislation

End of life management

- Registration by 'producers' in the Member States
- Extended producer responsibility:
 - finance and organise the separate collection and treatment of waste batteries
 - promote the separate collection of batteries
 - provide information to end-users and report to the competent authority
 - role of producer responsibility organisations
- Collection rates, recycling efficiencies (weight) and material recovery targets (Co, Cu, Pb, Li, Ni)
- Facilitating second life: repurposing, remanufacturing

Gradual application

- Some requirements started to apply on 18 February / 18 August 2024, mainly
 - Restrictions/markings on hazardous substances
 - Safety of stationary battery energy storage systems
 - Reporting on performance and durability parameters (EV, industrial and LMT batteries)
 - (restricted) Access to state of health parameters in the battery management system
- Some others will apply after the Commission has adopted detailed rules, e.g.
 - Carbon footprint (work on calculation rules nearly finished)
 - Thresholds for performance and durability (industrial and LMT batteries)
 - Recycled content

Life cycle carbon footprint

- Applicable to electric vehicle, industrial (above 2 kWh), and light means of transport batteries, *with different timelines*
- Staged approach: first declaration of the carbon footprint, then a comparative label, and lastly maximum thresholds
- Per battery model, per manufacturing plant, calculated as kg CO₂-eq. per kWh of the total energy provided by the battery over its expected service life
- Methodology based on Product Environmental Footprint, to be detailed by the Commission
- Draft methodology for electric vehicle batteries published:

https://ec.europa.eu/info/law/better-regulation/have-your-say/initiatives/13877-Batteries-for-electric-vehicles-carbon-footprint-methodology_en

Performance and durability

- Reporting from August 2024 for EV, industrial (>2 kWh) and LMT batteries
- Parameters:
 - Rated capacity (in Ah) and capacity fade (in %)
 - Power (in W) and power fade (in %)
 - Internal resistance (in Ω) and internal resistance increase (in %)
 - Where applicable, energy round trip efficiency and its fade (in %)
 - The expected life-time of the battery under the reference conditions for which it has been designed, in terms of cycles, except for non-cycle applications, and calendar years
- Standardisation on measurement methods ongoing
- Commission will set thresholds – only for industrial and LMT batteries, to apply from 2027/2028

State of health parameters

- From August 2024: state of health parameter(s) in the battery management system, accessible by the owner, for:
 - Evaluating the residual value or remaining lifetime
 - Second-life evaluation
 - Participating in the electricity market
- Parameter for EV batteries:
 - state of certified energy (SOCE)
- Parameters for stationary battery energy storage system and LMT batteries:
 - remaining capacity, remaining power capability, remaining round trip efficiency, evolution of self-discharging rates, the ohmic resistance (where possible)
- Standardisation on measurement methods ongoing - for industrial and LMT batteries

Battery passport

- From February 2027, for EV, industrial (>2 kWh) and LMT batteries, accessible via QR code:
 - Public information on the battery and sustainability
 - Information for dismantling, remanufacturing, recycling, including state of health – for those with legitimate interest; access rights still to be specified
 - Compliance information for authorities
- Decentralized, technical operation developed jointly with the digital product passport, through standardisation:
 - https://ec.europa.eu/growth/tools-databases/enorm/mandate/604_en

Thank you